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RESEARCH ARTICLE

Anomaly Detection on Smart Grids With Optimized Convolutional Long Short-Term Memory Model

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ABSTRACT The integration of digital technologies into traditional power systems has increased the efficiency and sustainability of power grids, transforming traditional grids into smart grids. However, this transformation has also introduced new vulnerabilities, such as susceptibility to false data injection (FDI) attacks, which can lead to significant energy theft. Recent reports estimate that these attacks cost utility providers approximately 101 billion dollars annually. This study presents an approach for anomaly detection in smart grids through energy consumption readings from smart meters on the customer side using an optimized lightweight convolutional long short-term memory (ConvLSTM) model. This study benchmarks and evaluates different machine learning models against seven FDI attacks, which are multi-class labeled. The evaluated machine learning models include traditional shallow detectors, deep learning-based detectors, and hybrid models that employ both horizontal and vertical detection strategies. Through extensive experimentation, the optimized ConvLSTM model is shown to demonstrate superior performance in detecting attacks; it achieves a high accuracy of 91.3% compared with other models in classifying these attacks. The results indicate that the proposed model provides a robust solution for improving the security and reliability of smart grids, and it offers significant benefits to utility providers who seek to mitigate energy theft and enhance grid resilience.

INDEX TERMS Anomaly detection, energy consumption, energy theft detection, operational technology, smart grid.

I. INTRODUCTION

Smart grids (SG) are the next generation of power grids for producing, transmitting, and distributing power. Households can now be integrated with an intelligent network of sensors that use the Internet of Things (IoT) through 5G networks to enhance the automation, efficiency, and sustainability of the grid. Unlike the traditional power grids that depend on the unidirectional flow of power from the point of generation to the point of consumption, households are now able to produce their own power using renewable energy and to push any

excess power back onto the grid [1]. This is facilitated by the advanced meter infrastructure (AMI), which comprises smart meters supported by two-way communication between the smart meter and grid [2]. This adoption reduces the dependency of households on fossil fuel and diversifies the sources of energy, making the grid resilient in terms of distribution. Although the introduction of smart grids enhances the efficiency and sustainability of the power grid, there are vulnerabilities and risks associated with any new technologies that affect its reliability and integrity. The complexity of the communication infrastructure between the AMI and the production side or power distribution make communication vulnerable. In the case of the smart grid, the

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risks mainly involve the following: first, data management for billing and protecting personally identifiable information (PII); second, smart grid security against cyber threats such as DDos attacks that make the service unavailable and ransomware attacks that compromise the availability of the grid [49], [50]; and third, the system complexity and integration challenges of various technologies, including sensors, smart meters, and distributed energy resources. These risks add a layer of complexity to smart grid systems. This complexity may result in operational inefficiencies, especially if the standardization between different devices and systems is lacking. In terms of data management, the operation of smart grids involves the collection and transmission of large amounts of data, including real-time information about energy consumption patterns. Although these data are essential to optimizing grid performance, the operation of smart grids raises significant concerns regarding data privacy due to the fact it involves PII. Also, the increase in cyber threats that target the grid, including grid disruption, false data injection (FDI), energy theft, and making the grid unavailable for consumers, poses significant challenges pertaining to smart grid security.

To address these challenges, this study aims to address the security aspects of the grid, specifically reducing the risk of FDI. FDI happens from the consumer side, such as when a malicious consumer manipulates the smart meter by adjusting energy consumption readings and state emission readings.

Injecting data that misrepresents reality is a common practice that is done either to abuse the payment system for gain, such as maliciously reporting that the consumer has generated electricity through photovoltaic panels or wind turbines to be compensated financially, or to cause a loss of profit to the energy service provider [3]. A study published in 2023 estimated that power companies incur 101 billion dollars in losses each year due to energy theft and fraud [4]. Real-life examples have occurred in India, where millions of dollars were lost due to the bypassing of smart meters [41]. It was also reported that in Mexico, authorities have cracked down on individuals who tamper with smart meters [42].

To address this problem, various anomaly detection techniques used in information technology (IT) have been proposed [47], and have evolved into different lines of products ranging from application/network anomaly detection, which is the extended detection and response (XDR) platform that correlates multiple data sources in order to detect malicious behavior, intrusion detection systems (IDSs), which detect unauthorized access, and intrusion prevention systems (IDSs), which block unauthorized access, and finally the user entity behavior analytic (UEBA) platform, which creates a baseline for the users within the environment from their past behaviors and flags any anomalous behavior. However, there were no efforts to address operation technology (OT), which resulted in major security issues, especially in IoT networks and smart grids. Because OT telemetry

differs depending on the application, this study focuses on energy theft detection by combining machine learning models. Moreover, it evaluates the performance of different models in various attack scenarios while taking into account the time factor and its partial reduction to perform the attack.

A. CONTRIBUTION

In this study, we propose a convolutional long short-term memory (ConvLSTM) model, addressing the limitations in electricity theft detection (ETD) by introducing the following key contributions:

- **Proposed Vertical Hybrid Architecture:** We develop a lightweight, vertical hybrid model that integrates a convolutional neural network (CNN) with long short-term memory (LSTM) cells into a unified architecture. Unlike cascaded hybrid models that process components sequentially, the proposed vertical structure ensures efficient feature extraction and sequence preservation within a single-stage design. This reduces computational overhead and improves classification accuracy for multi-class datasets.
- **Extension to Multi-Class Classification:** While most prior ETD research focuses on binary classification (e.g., theft vs. non-theft), this study extends the scope to multi-class classification. Our approach not only identifies attacks but also classifies attack types, enabling the utility provider of understanding anomalous behavior in electricity consumption. This approach aligns with real-world requirements, where multiple attack types must be detected and categorized simultaneously.
- **Comprehensive Baseline Comparisons:** We evaluate the proposed ConvLSTM model against a wide range of baseline models, including: Shallow classifiers: Random Forest (RF) [35] and Extreme Gradient Boost (XGBoost) [36]. Deep learning models: LSTM and autoencoders (AE) with neural network (NN) and CNN backbones. Alternative hybrid vertical models: A CNN-GRU hybrid for comparison with the ConvLSTM. These comparisons provide a thorough analysis of model performance across various architectures, demonstrating the advantages of our method.

B. ORGANIZATION

This manuscript covers previous work conducted in the field of anomaly detection II and is structured as follows. Section III gives an extensive overview of the proposed model and the problem statement. Section V covers the dataset and the induction of attacks within. In Section IV, the manuscript highlights the ConvLSTM model, and in Section VI, a comparative approach that evaluates different models is presented. Section VI-F explores the results of the model evaluations. Finally, the manuscript concludes in VII by highlighting future directions.

II. RELATED WORK

A review of the literature on existing anomaly detection within smart grids revealed that this domain is still immature relative to the work that has been done in the field of anomaly detection in conventional IT environments. Zhang et al. surveyed time series anomaly detection in smart grids while focusing on shallow machine learning (ML) techniques such as simple supervised, unsupervised, and simple NNs [9]. Based on their results, the authors identified three potential avenues of research: creating anomaly benchmarks for time series in smart grids, learning transfer for anomaly identification, and enhancement techniques for online anomaly detection relative to current offline trained models.

Moreover, Alkuwari et al. surveyed the above techniques with contextualized attack vectors and data sources [10], and they found that the use of hybrid model—a “combination of two models”—was more effective for detecting anomalies in smart grids than the use of a single model. Several studies have addressed the detection of FDI in simple shallow classifiers such as support vector machine (SVM) [7], RF [6] and deep neural architecture such as the CNN [8]. Deep neural networks (DNNs) such as the autoregressive integrated moving average (ARIMA) were also used by Chou et al., and gave an anomaly prediction with an accuracy of 86%–96% accuracy [31]. Recently, more advanced detection models and hybrid models have shown deep analytic capabilities for anomaly detection. Elhadad et al. used a stacked model, which comprised RF to label the data as well as decision trees (DT), as a final prediction node that achieved 90% accuracy [16]. Moreover, Hu et al. proposed a combined Deep Siamese Network (DSN) and DNN that offered a 95% detection rate [14]. Another work proposed a CNN combined with a GRU that yielded an accuracy of approximately 89%. In that work, the CNN targeted data feature selection, whereas the GRU was used for the classification of data [12]. Takiddin et al. proposed a complex architecture in which an AE with an embedded LSTM and CNN was used as the final classifier node. The model yielded a detection rate of 96% in a white-box setup [17]. Moreover, Zheng et al. used AEs with embedded recurrent neural networks (RNNs) on input time series consumption data in California. Their findings showed that anomalies could be successfully captured by the suggested encoder-decoder structure with a precision greater than 95% [52].

As clearly shown from the reviewed research conducted in this field, hybrid models, where different ML algorithms can be stacked and manipulated to gain the most effective combination, have yet not been sufficiently leveraged. More importantly, the reported models were evaluated against conventional attacks that were binary-labeled as malicious or benign. Specifically, hybrid models need to be leveraged to profile the energy consumption of users and to classify different types of malicious attacks and types of benign users based on this profiling.

III. SYSTEM OVERVIEW AND PROBLEM DESCRIPTION

A. SYSTEM MODEL

In a typical SG environment, smart meters are deployed at residential or commercial sites to track energy consumption over predefined intervals, such as every hour or every day. These smart meters transmit data to a central utility server, which processes the consumption data for the purpose of billing and monitoring. Under normal circumstances, the consumption data reported by the meters reflects the actual energy used by consumers. However, malicious users are able to manipulate this data, either through physical tampering with the smart meter or by intercepting and altering the digital transmission of the consumption records. The main goal of the malicious user is to reduce the quantity of energy consumed, as recorded by the utility, thereby reducing their electricity bill. These manipulations have significant economic implications and may cause revenue loss for utility companies and affect grid reliability. This problem is exacerbated in advanced SGs, where smart meters automatically record and report consumption data in near real time. Although these systems are sophisticated, they are vulnerable to tampering and exploitation by malicious actors.

B. PROBLEM DESCRIPTION

Electricity theft detection in the consumption domain poses several significant challenges that make it a difficult problem to solve effectively. One of the main issues is the variability in the legitimate patterns of electricity usage, which can fluctuate according to a range of factors, including household size, seasonal weather conditions, and daily routines. This natural variability makes it difficult to establish a clear baseline for normal consumption and increases the risk of both false positives (wherein legitimate fluctuations are flagged as theft) and false negatives (wherein actual theft goes undetected). Additionally, attackers often exploit natural fluctuations by introducing small, gradual changes into their reported consumption. These small perturbations are designed to prevent detection by blending in with legitimate variations in the data. For example, a malicious customer might reduce their reported consumption by only 5%–10% during peak usage hours, or selectively bypass the meter for short periods. Traditional methods have difficulty detecting such subtle manipulations, as they typically rely on threshold-based or rule-based systems that are prone to generating a high number of false positives in response to small perturbations, as they cannot effectively differentiate between normal fluctuations and malicious behavior. Furthermore, electricity consumption data are inherently time-dependent, meaning that electricity usage at a given point in time is often correlated with usage at previous or subsequent time points. For example, electricity usage may follow a daily cycle, where consumption peaks during certain hours (e.g., evening) and drops during others (e.g., in the early morning). This temporal dependency adds a layer of complexity to

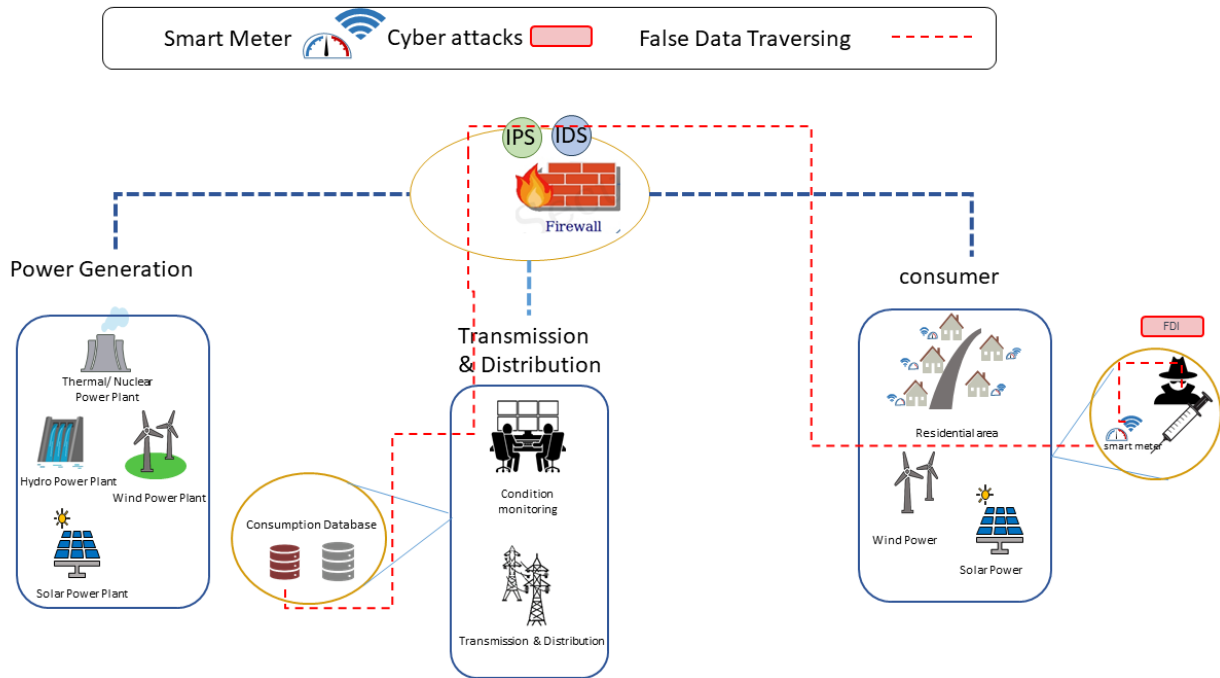


FIGURE 1. Graphical representation of attack scenario within SG.

the problem, as any detection system must be capable of analyzing sequential data to accurately identify deviations from expected patterns. Another significant challenge arises from the diverse attack techniques employed by malicious customers. Some attackers may use a fixed reduction in their reported consumption throughout the day, whereas others may use variable reductions that change hourly. More sophisticated attackers might target specific time periods, such as peak usage hours, to reduce the cost of their bill while avoiding detection during off-peak times. In other cases, attackers may bypass the smart meter entirely for short periods and then report zero consumption for those periods. Each of these attack types presents unique challenges for detection systems, as the systems must be able to recognize a wide range of manipulation strategies.

C. THREAT MODEL

There are several threat models that incorporate and correlate the cyber physical threat such as MITRE ATT&CK [25] which list all the possible Techniques, Tactics and Procedures that a threat actor might take to accomplish his objective in the Operational technology realm. The diamond model [30] on the other hand, study the intent and capability of the threat actor, the infrastructure required to preform the attack and finally the victim and its defense capability. In this study, the reference of the threat model is based on Advanced Cyber-Physical Power System (ACPPS) kill chain proposed in [48], the framework examines the cyber-physical APT stages on power grids through attack stages for a thorough analysis of Advance Persistent Threat (APT) on power systems that

cause cascading failures and a blackouts. In attack stage D of the ACPPS, it is assumed that the adversary already has access to the OT device in our study this is reflected by the adversary access to the smart meter. The sub-stage number 14 highlights the unauthorized commands induced by the adversary, in this study it indicates the FDI preformed by the attacker as highlighted in Figure 1. Finally, stage F indicates the impact of the attack, the sub-stage number 22 highlights the financial impact that results from the cyber attack. Additionally, this study consider the threat model as a blackbox threat model scenario. In this setup, the attacker has no idea about the detection mechanism or the model, or parameters involved. This mimics a real-life scenario where the attackers have no visibility or knowledge of the ML models.

IV. PROPOSED ANOMALY DETECTION MODEL

The proposed detection model is hybrid; hence, it is not a traditional model that is layered in a sequential form as in existing literature. The proposed model is based on a CovLSTM model that is characterized by its hybrid ability in providing the vertical concept such that the main container of the CNN is embedded with the LSTM cells. This is contrary to the case of a normal CNN architecture, where it is embedded with simple neurons [43].

A. CONVOLUTIONAL NEURAL NETWORKS

Traditionally, CNNs are mainly used for video, image, and audio processing due to the nature of models that learn features from spatial and time dimensions. This is achieved

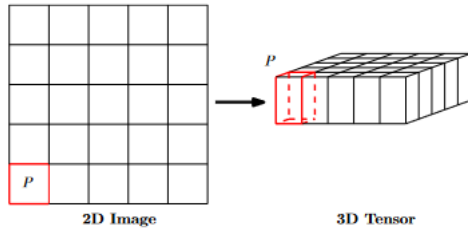


FIGURE 2. ConvLSTM transformation of 2D image to 3D tensor.

using a kernel operation, where a sub-matrix captures features through a convolutional operation on the input. In this context, the CNN captures the features from user consumption data, which contains both numerical and time values.

B. LSTM

The concept of LSTM was introduced in the late 90s and is commonly used for natural language processing (NLP) [15]. They are considered to be RNNs, and they solve the vanishing gradient problem of regular RNNs. LSTM was selected in for the ETD problem due to its ability to capture deep relations between each sequential h_{t-1} and the input at each time x_t , forget gate f_t , input gate i_t , and to finally produce the output gate o_t .

- $i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$
- $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$
- $q_t = \tanh(W_q \cdot [h_{t-1}, x_t] + b_q)$
- $o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$
- $c_t = f_t \odot c_{t-1} + i_t \odot q_t$
- $h_t = o_t \odot \tanh c_t$

C. ConvLSTM

In combining the two models, the architecture is as follows: the main container is the CNN, where the maxpooling layer is replaced by an LSTM that captures the deep relations between the time series consumption data. In the LSTM inner gates, the matrix multiplication state is replaced by convolutional computation in the fully connected layers [43]. In the following equations, (*) denotes the convolution operation, and (·) denotes the Hadamard product. Normally, the CNN accepts data as a 2D image, but in the ConvLSTM, it is taken to be a 3D tensor, as shown in Figure 2.

$$i_t = \sigma(W_{x_i} * X_t + W_{h_i} * H_{t-1} + W_{c_i} \cdot C_{[t-1]} + b_i) \quad (1)$$

$$f_t = \sigma(W_{x_f} * X_t + W_{h_f} * H_{t-1} + W_{c_f} \cdot C_{[t-1]} + b_f) \quad (2)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tanh(W_{x_c} * X_t + W_{h_c} * H_{t-1} + b_c) \quad (3)$$

$$o_t = \sigma(W_{x_o} * X_t + W_{h_o} * H_{t-1} + W_{c_o} \cdot C_t + b_o) \quad (4)$$

$$H_t = o_t \cdot \tanh C_t \quad (5)$$

D. ConvLSTM MODEL CREATION

This study uses the PyTorch Library to construct the ConvLSTM, which enables one to create and modify the

models with greater customization [44]. As stated above, the main container model is a CNN in which the first layer is the convolution layer. A reshaping layer is introduced to reshape the tensor from 2D to 3D, as the input data are 2D. The LSTM neurons are embedded within the CNN model. These neurons (128 neurons) are fully connected. The hidden layers are considered to be hyperparameters and will be discussed in the optimization section VI-B. The output layer is also comprised another fully connected layer with 512 neurons and a softmax function. The reshaping layer is the final output of the classification operation.

V. DATASET

In this study, the publicly available dataset from the Irish Smart Energy Trail (ISET) is used [46]. The dataset contains energy consumption data collected at 30 minute intervals in KWh/user. The collection covers more than 3000 households spanning a period of 18 months. Figure 4 shows the trend of average energy consumption at each half hour of the day over the span of one month for multiple users. The figure shows clearly different usage patterns specifically in the duration of work hours 8AM to 4PM the energy consumption drops, which is predictable due to the fact that people tend to be outside the household in this duration. Figure 5 shows how attacks has drastically changed the energy consumption pattern of the same users. The majority of the consumption now is reported in the duration of work highlight above and the reported consumption is a fraction of what is really consumed. The energy consumption is modeled as $\mathbf{U}(\mathbf{E}, \mathbf{t})$ where $\mathbf{U}$ is the user, $\mathbf{E}$ is the energy consumption, and $\mathbf{t}$ is the time. The dataset contains 1.5 million entries, as described in Table 1.

A. DATA PROCESSING

The dataset underwent preprocessing that included the elimination of the outliers, which lay in the last quantiles of the data, and any other deviation in the dataset. A dictionary of Meter IDs was created to couple each Meter ID with its readings throughout the period of 18 months. A moving average window was used to generate 48 readings per Meter ID that represented the 30-minute intervals used for creating user profiles throughout the 24 hour day. These readings were tied to each unique Meter ID.

B. ATTACK METHODS

The proposed attacks that were applied to the dataset are described in this subsection. The attacks are intended to introduce false readings of the energy consumption data. These false readings are combined with the original data to produce the malicious and the benign datasets. The attacks that are usually used fall into two main categories: (1) conventional attacks that use probabilities and time to manipulate the injected false data (mainly derived from [19]), and adversarial attacks that use ML algorithms to produce perpetuated

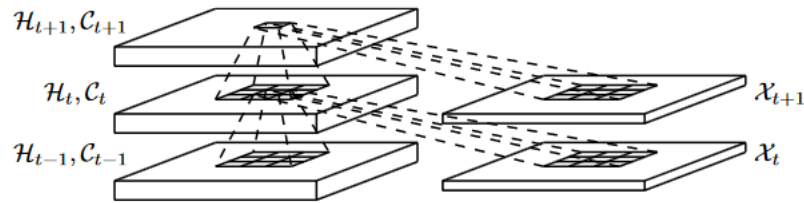


FIGURE 3. Inner structure of ConvLSTM.

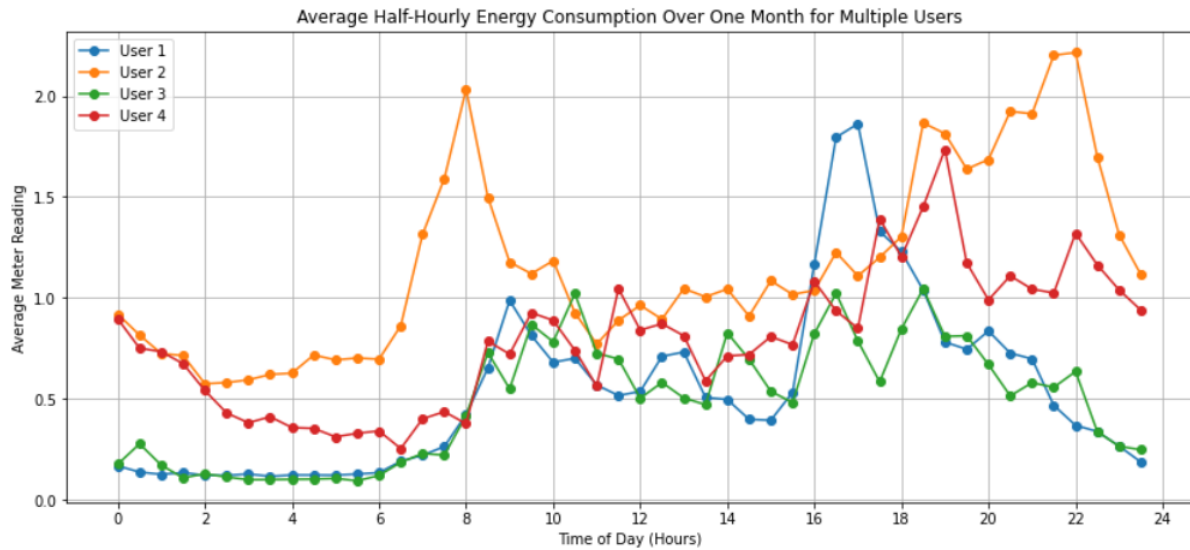


FIGURE 4. The average power consumption for multiple users over one month.

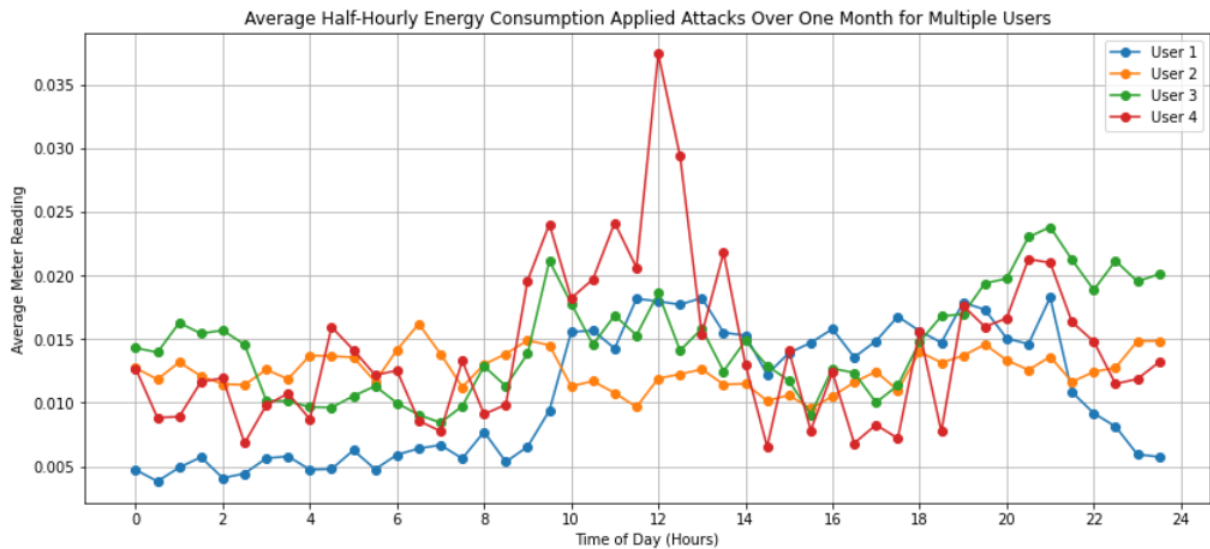


FIGURE 5. The average power consumption for multiple users after attack applied over one month.

samples to be injected into the data. This research is focused on the first type of attack. Below are descriptions of attack

types that use various functional strategies. These attacks target different dimensions of energy data manipulation,

TABLE 1. Description of ISET dataset.

Dataset Description			
Dataset	Duration	Type of Class	Number of Samples
ISET Dataset	18 Months	Benign - One Class	1.5 Million
Induced Attack Functions (7)	-	Malicious – Multi-Class (1–7)	1.5 Million for each
Total Samples			12 Million

including partial reduction, selective bypassing, and price-based control methods. The mathematical formulations given illustrate each attack function's methodology and impact on energy consumption data $U(E, t)$, where E represents energy consumption and t denotes time.

1) PARTIAL REDUCTION

The reported energy consumption is consistently reduced and could be one of the following:

- *Constant Multiplication*: This attack reduces reported consumption by multiplying the energy reading by a constant factor, α , where $\alpha < 1$. Mathematically, it is represented as $f_1 U(E, t) = \alpha \cdot U(E, t)$. This reduction uniformly scales down the readings, thereby potentially allowing consistent under-reporting over time.
- *Variable Multiplication*: With this attack, the energy consumption is scaled by a variable factor $\beta(E, t)$ which varies temporally to allow the data to be manipulated with greater flexibility and sophistication. This attack is represented by $f_2 U(E, t) = \beta(E, t) \cdot U(E, t)$, where $\beta(E, t) < 1$.

2) SELECTIVE BYPASS

This attack strategy introduces periods during which the reported energy consumption is set to zero, while at other times, the actual consumption value is recorded. This selective manipulation is defined as:

$$f_3 U(E, t) = \begin{cases} 0, & \text{if } t \in [t_x, t_{x+n}] \\ U(E, t), & \text{otherwise} \end{cases}$$

This bypassing technique can mimic intermittent device failures and maintenance, which makes its detection quite difficult.

3) PRICE-BASED CONTROL

The attacker exploits demand patterns by adjusting reported values based on fixed or dynamic pricing conditions

- *Fixed Constant Reporting*: The energy consumption is averaged across a time frame, and this average value $\bar{E}_n$ is consistently reported during the day. Mathematically, $f_4 U(E, t) = \bar{E}_n$, which creates a baseline value that could mask real-time consumption spikes.
- *Dynamic Variable Reporting*: Similar to partial reduction with a variable multiplier, this function manipulates the reported values using a time-varying function

$\beta(E, t)$, thereby ensuring that reported values remain within a controlled range. This function is represented as $f_5 U(E, t) = \beta(E, t) \cdot U(E, t)$.

- *Shifted Reporting*: During periods of high demand, this attack shifts data by a predetermined interval T , so as to present an earlier or lower value to evade peak-demand penalties. The shifted function is represented by $f_6 U(E, t) = U(E, t - T)$.
- *Swapped Extremes*: This attack function swaps high values with low values, thereby creating an artificial balance in the data to smooth out consumption peaks and avoid detection. It is mathematically represented as $f_7 U(E, t) = U(E_{\max}, t) \oplus U(E_{\min}, t)$.

The datasets that have eight classes are normalized by the sklearn library [26] and then split into training and testing tests using the same library.

VI. EVALUATION AND BENCHMARKING

This section describes the parameters of the environment, training, and testing. The entire experiment was conducted on a Ubuntu Linux operating system that had two graphic processing units on Nvidia GTX 4090 cards, each having 10 GB of memory, and 64 GB of RAM.

A. TRAINING OF DEEP CLASSIFIERS

Due to the enormous amount of data (17 million entries), the machine hardware (processor and memory) could not manage the load. Thus, a data generator was used throughout the training [39]. The data generator made it possible to create multiple datasets in real time to be fed into the GPUs in a parallel manner. The generator was employed to efficiently load data in batches, addressing the challenge of handling large datasets that could not fit into memory during training. The input to the generator consisted of file paths or indices for the raw data on disk, while the output consisted of processed batches of input features and labels, ready for the ConvLSTM model. The training was run on 300 samples per epoch. The use of the generator optimized memory usage and computational efficiency without compromising the data integrity. The testing phase of the experiment utilized the same data generator to evaluate its performance on data by leveraging a systematic approach involving batch-wise data loading. The test data is processed using a data loader, ensuring efficient handling of large datasets by dividing them into manageable batches. The testing phase uses an accuracy metric that provides a comprehensive assessment of the model's effectiveness in multi-class classification. This

TABLE 2. Induced conventional attack functions.

Attack Type	Method	Mathematical Representation
Partial Reduction	Multiplied with a constant	$f_1 U(E, t) = \alpha \cdot U(E, t)$, where $(\alpha < 1)$
Partial Reduction	Multiplied with a variable	$f_2 U(E, t) = \beta \cdot U(E, t)$, where $(\beta(E, t) < 1)$
Selective Bypass	Changes to zero during a selective period of time; actual value is reported during other periods	$f_3 U(E, t) = \begin{cases} 0, & \text{if } t_x - t_{(x+n)} \\ U(E, t), & \text{elsewhere} \end{cases}$
Price Base Control	Reporting a fixed constant throughout the day from an average of the consumption values	$f_4 U(E, t) = \bar{E}_n$
Price Base Control	Reporting a dynamic variable	$f_5 U(E, t) = \beta \cdot U(E, t)$, where $\beta(E, t)$ is a fraction of $U(E, t)$
Price Base Control	Reporting a lower value when the demand is high by shifting data to the left	$f_6 U(E, t) = U(E, t - T)$
Price Base Control	Swapping highest reported values with the lowest values	$f_7 U(E, t) = U(E_{\max}, t) \oplus U(E_{\min}, t)$

phase highlights the model's ability to generalize across diverse attack types, emphasizing its suitability for real-world applications.

B. OPTIMIZATION OF THE MODEL

An optimization function was created to further adjust the weights of the parameters. The Optuna library was utilized on the hybrid vertical models, i.e., ConvLSTM and CNN-GRU [51]. Optuna enables model optimization through the creation of multiple trails and the testing of different combinations of hyperparameters, such as learning rate, number of LSTM layers, number of hidden layers, CNN channels, and kernel size. In this study, 50 epochs were used in 50 trials, as shown in Table 4. The optimizer used was the Adam optimizer, and the loss function was binary cross entropy loss. The end results for the ConvLSTM hyperparameters are shown in Table 3.

In this work, accuracy was used as the main evaluation metric for the different selected ML models, whether for training or testing purposes, because all the datasets were labeled as benign data = (0) or malicious data = (1–7) to classify each attack type. At the end of each neuron of the ML models, a classification occurs as either $y = (1-7)$ for malicious classification or as $y = (0)$ for benign classification. Hence, an accuracy metric would check the number of correctly classified samples against the total number of samples multiplied by 100 to obtain the percentage

TABLE 3. Optimized ConvLSTM Hyperparameters.

Optimized Model Hyperparameters	
Hyperparameter	Value
Learning Rate	0.0001
Number of LSTMs	3
Number of Hidden Layers	512
CNN Channels	128
Kernel Size	5

of classifications that are accurate. Additionally this work utilized metrics such as precision, which investigates the ratio of true positive predictions to the total predicted positives (true and false positives); recall that is the ratio of true positive predictions to the total actual positives. Lastly, this work used the F1 score, which takes into consideration both precision and recall.

C. BENCHMARKING

For the purpose of creating a baseline that is able to compare the proposed model with other known models, the proposed model is evaluated, in terms of accuracy, against

Algorithm 1 ConvLSTM Training

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1: Input: Energy Consumption Data Per User  $E_t$ 
2: Output: Benign/Malicious Classification/Attack Type
3: for each  $X \in E_t$  in each epoch do
4:   At each ConvLSTM layer
5:   Compute:
6:   for Each hidden layer  $l=1,2,..L/2$  do
7:     for Each time step  $t$  do
8:       Input gate activation using function 1
9:       Forget gate activation using function 2
10:      Output gate activation using 4
11:     end for
12:   end for
13: end for

```

four other models under the same threat conditions, which are the blackbox attack and the seven conventional attacks. The two shallow classifiers against which the proposed model is evaluated are the RF and XGBoost classifiers. The deep neural classifiers against which the proposed model is evaluated are LSTM, AE-NN, AE- CNN, and CNN-GRU in a vertical hybrid model. The approach that is used in splitting the data is consistent throughout the phase of benchmarking and evaluating the models. It is based on using 80% of the data for training and using the remaining 20% for testing [40]. The 80/20 training/testing split selected to ensure the ConvLSTM model effectively captures both spatial and temporal dependencies while training and maintaining a meaningful evaluation set. Given that ConvLSTM models rely on sequential data, a larger training set allows the model to learn patterns over time, reducing the risk of poor generalization. In addition, the optimization process in the training phase involves iterative updating of the model parameters to minimize the loss function. In each training iteration, the model predicts the output label based on the input, and the loss is computed by comparing the predicted outputs to the actual labels. Using back propagation, gradients of the loss function with respect to the model parameters are calculated. The Adam optimizer is utilized to update the model weights, ensuring efficient learning by dynamically adjusting the learning rate for each parameter. When cross-validation is applied, the optimization resets for each fold, ensuring that the model is trained separately for each validation subset. This iterative optimization strategy allows the model to progressively minimize errors and improve performance over time.

D. SHALLOW CLASSIFIERS

For the shallow classifiers, the lazy predict library, which consists of 30 classification algorithms, was used [34]. The aim of using this library is to create a baseline for the shallow classifiers that the advanced ones could be compared with. For the purpose of utilization and applicability, this study used the two best-performing algorithms out of

TABLE 4. Different combinations of hyperparameters used in 50 trials.

Different Combinations of Hyperparameters Used in 50 Trials	
Hyperparameter	Values
Batch Size	200
Learning Rate	$1e^{-5}$, $1e^{-3}$
Number of LSTM Layers	1, 3
Number of Hidden Layers	128, 256, 512
CNN Output Channels	32, 64, 128
Kernel Size	3, 5, 7

30 algorithms. RF and XGBoost were found to be the two best-performing models.

E. DEEP CLASSIFIERS

The deep classifiers were taken from the Keras Library [38]. Keras was chosen for its simplicity, yet it was considered to be a powerful tool that enabled the creation of different types of models, the manipulation of parameters, and the weighting of the activation functions for the evaluation of different models.

1) LSTM

In this study, an LSTM based on [15] was created with 128 neurons and 3 hidden layers.

2) AE-LSTM BASED

In this section, a detailed overview of the baseline component of all the hybrid horizontal classifiers is dissected. The AEs embedded with LSTM neurons differ from normal AEs with embedded NNs by their a temporalized relation between each reading of energy consumption, which captures the time dimension and creates a relationship with the previous reading. This helps to create a more accurate representation of the features. The model consists of four stacked LSTM layers, where the first two act as an encoder, creating a compact representation of the data. The last two LSTM layers function as a decoder, expanding the representation back to the original input features dimensions. The forward feeding mechanism ensures the input is sequentially fed through these LSTM layers, maintaining the same shape as the input, allowing the model to learn an efficient representation of sequential patterns. This structure enables the AE-LSTM based to reconstruct normal patterns effectively.

3) AE-NN

The AEs embedded with the LSTM neurons, as explained above, create a temporalized relation between each reading of energy consumption to capture the time dimension and create a relationship with the previous reading. This is essential to creating a more accurate representation of the features. To accomplish this task, the AE-NN model, which was created based on the AE structure described above, is first constructed with four layers of LSTM neurons, with two layers as encoders and two layers as decoders as follows (64,

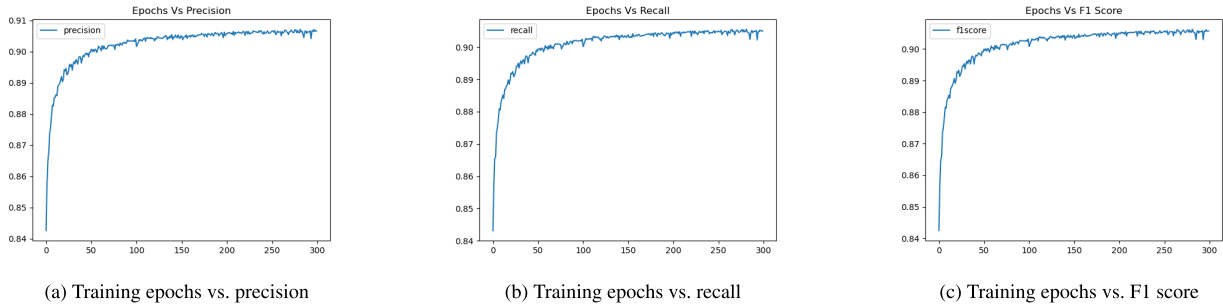


FIGURE 6. ConvLSTM training vs. precision, recall F1 score metrics.

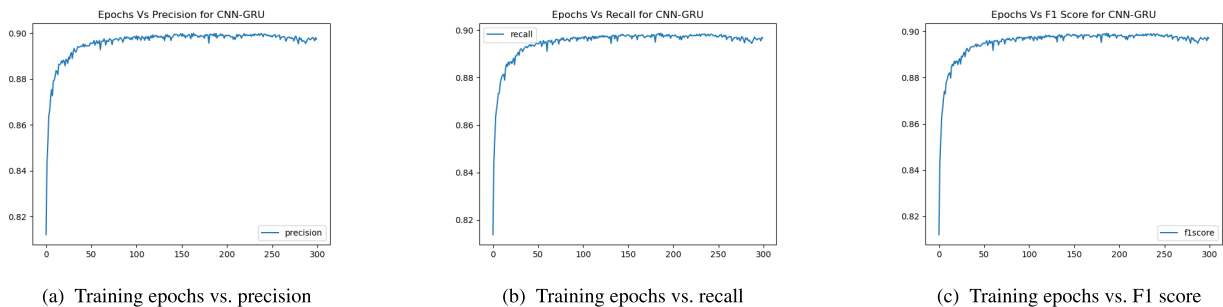


FIGURE 7. CNN-GRU training vs. precision, recall, and F1 score metrics.

32, 64, 32), respectively, for each layer. Rule, Adam, and Mean Square Error are used as the loss function.

After the decoded data are passed to the NN model from the AE, the structure of the model is as follows: four layers consist of neurons with structured as (288, 128, 16, 2) for each layer, respectively. However, with respect to the other parameters, Rule is used as the activation function, Adam is the optimizer, and the cross entropy loss function is the loss function. The training of the AE-NN model took place in two separate systems, meaning that training was conducted separately for the AE and the NN.

4) AE-CNNs

The same setup for the AE was used based on the description above of the AE model. With regard to the CNN model, the data first had to be reshaped due to the fact that the CNN was based on image processing. Thus, the data were transformed to 16×16 images at the input layer. The model also contained a maxpooling layer, another reshaping layer, a flattening layer, a dense layer, and finally a classification layer. Such a configuration was previously reported to achieve 96% accuracy [17]. It was thought that the model would perform much better in solving binary classification problems than in solving multi-class problems. Moreover, the model was trained as two separate models.

5) CNN-GRU

This model is introduced as another benchmark in our study of different combinations of vertical hybrid models such as

the proposed ConvLSTM that yielded the best accuracy for ETD. The CNN in this setup processes, through the reshaping layer, the features of consumption as an image, and the GRU is an evolved version of LSTM (RNN). The GRU nodes are based on two gates: the update gate and the reset gate.

F. RESULTS DISCUSSION

The shallow detectors, i.e., RF and XGBoost, showed the expected detection rates of 65% and 64%, respectively. This is expected because those detectors are not able to retrieve the inner relationships between the samples or to create a spatial time correlation. Although the LSTM is in the category of deep detectors, it did not achieve good results compared to shallow detectors, and it achieved an accuracy of 75.8%. It reaffirms that the combination of CNN based LSTM is able to draw a deeper relationship than the LSTM alone. This is due to the fact CNN through its architecture is able to extract the features and represent the spatial feature plane much accurately than the LSTM alone.

This study reveals that the AE-NN model demonstrated slightly better detection performance, achieving an accuracy of 88.3% compared to 87.1% for AE-CNN. This difference, though marginal, can be attributed to the structural and functional characteristics of the secondary components in each architecture. It is believed that due to the fact that the NN serves as a fully connected layer-based classifier that is well-suited for handling low-dimensional and less spatially complex data. The simplicity of the system and the relatively low dimensionality of the dataset in this study

Preformance Metrics for Vertical Hybrid Models

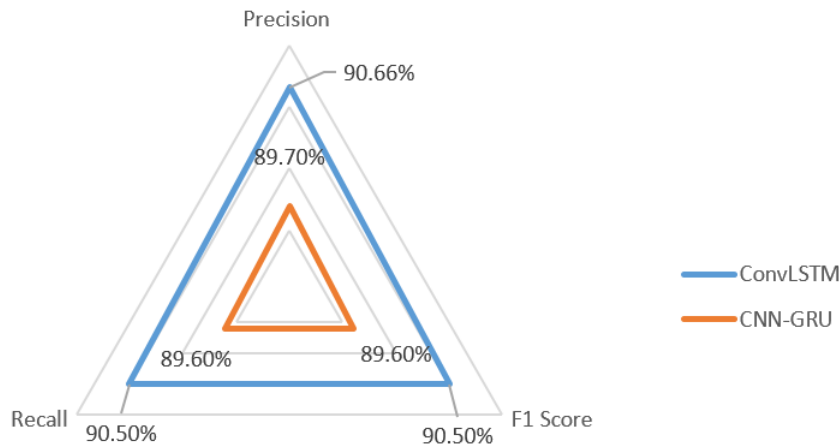


FIGURE 8. Training performance metrics for vertical hybrid models.

Accuracy Results per Model Category

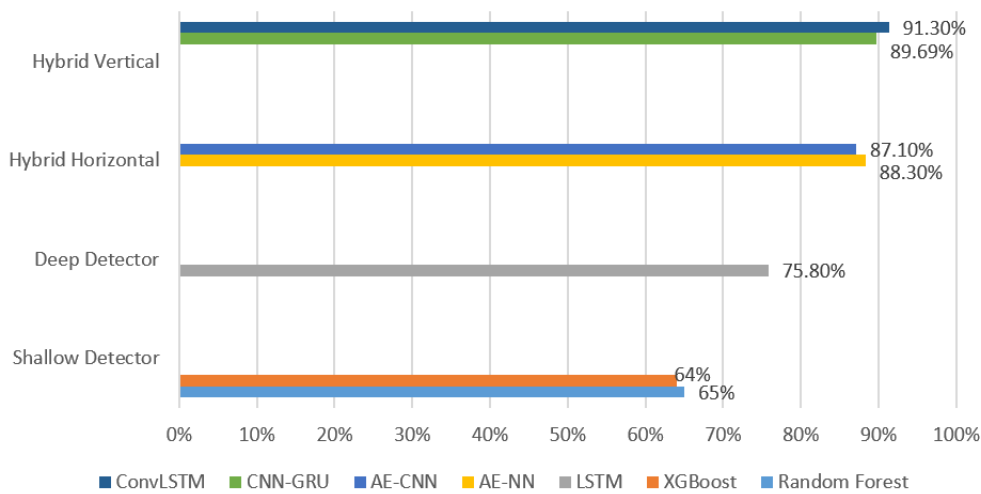


FIGURE 9. Accuracy results per model.

TABLE 5. Accuracy evaluation of different models.

Type of Classifier		Accuracy
Shallow Detector	Random Forest	85%
	XGBoost	84%
Deep Detector		
LSTM		75.80%
Hybrid - Horizontal	AE-NN	88.30%
	AE-CNN	87.10%
Hybrid - Vertical	CNN-GRU	89.69%
	The proposed ConvLSTM	91.30%

most probably allowed the NN component to effectively capture the relationships within the feature space. This

observation also highlights the importance of tailoring model architectures based on the nature of the problem and the type of dataset that the model would be handling rather than adopting more complex architectures by default.

To ensure a comprehensive evaluation of the proposed ConvLSTM model, we benchmarked against another vertical hybrid architecture namely the CNN-GRU model. This comparison aimed to validate the effectiveness of the proposed approach within the same experimental setup. Despite it showing promising results and achieved an accuracy of 89.69%. The model did not improve much from the accuracy it achieved. the model’s accuracy showed limited improvement, even after optimization with Optuna library.

To assess the model training performance and learning behavior of the hybrid vertical models, we used the metrics of precision, recall, and F1 Score. The results for the models are shown in Figure 8. Figures 6 and 7 show quite similar patterns for these metrics, which may indicate several things. First, the trends show an increase in learning for the first 100 epochs, which indicates that the model is effectively learning and exploring, and the LSTM component is building relations to explore the hidden patterns within the dataset. However, in the CNN-GRU, the exponential increase during the first 50 epochs shows rapid learning behavior, as the GRU has fewer parameters to compute than does the LSTM. Second, the models are processing a balanced dataset, meaning that one class will not be favored over another. Third, similar trends indicate that the models are robust in handling the spatial and temporal dimensions of the data. Lastly, from Figure 6 the convergence point of the precision, recall, and F1 score occurs in epoch number 200, which indicates the performance of the model has reached stability at this point; however, in the CNN-GRU model, the convergence point is at 100 epochs due to the lesser number of parameters. It is apparent from the results above and as displayed in Figure 9, that the proposed ConvLSTM model outperforms different models in detecting energy theft in a multi-class situation and is able to classify and label each attack type at 91.3% accuracy. Moreover, the ConvLSTM model achieved a precision of 90.6% during training, which indicates the model is able to minimize false positives, which makes the model more efficient when a security analyst is using it. The recall of the model is 90.5%, which suggests that the model is able to detect the majority of true positives and label them according to their respective attack labels. As the model achieved an F1 score of 90.5%, this indicates a strong balance between precision and recall and suggests that the model has high performance.

VII. CONCLUSION

Smart grids are essential for efficient modern energy distribution; however smart grids are known to be susceptible to various threats pertaining to data management and overall security. These vulnerabilities expose the system to significant risks, particularly from cyber threats that target sensitive data and operational stability. In this study, smart grid data were systematically subject to attacks to understand and counteract modern threat vectors that could compromise security systems. To simulate real-world attack conditions, seven different attack scenarios were created within a blackbox threat model based on the ACPPS framework to ensure the adversary actions are captured in a structural manner. These attack scenarios were designed to be realistic, and they focused on energy theft in particular, as this is an issue of growing concern in smart grid security. Each of the scenarios represented a unique type of attack, which were labeled as Attacks 1–7 and categorized as a multi-classification problem. This classification allowed for the

precise differentiation of threat types, aiding in targeted detection and mitigation strategies.

This research addresses an increasing demand for robust monitoring and protection mechanisms within smart grid systems. To address this need, a proposed lightweight and optimized model based on an ConvLSTM model is introduced. The proposed model is designed specifically to tackle energy theft detection with a high degree of accuracy. The model achieved an impressive 91.3% accuracy and was benchmarked against various ML models, which included the shallow, traditional ML detectors RF and XGBoost, as well as deep detectors like LSTM and advanced hybrid models with horizontal and vertical configurations, including AE-NN, AE-CNN, and CNN-GRU. Notably, the ConvLSTM model demonstrated advanced capability in distinguishing between the seven attack types. This success indicates the model's ability to learn and leverage the deep relationships between spatial and temporal dependencies within the data and to enhance detection precision.

Future research should explore the robustness of the ConvLSTM model when subjected to adversarial attacks, further ensuring its reliability in velocity data environments. Additionally, a study of the model's performance under such conditions will provide insights into potential vulnerabilities and the overall resilience of smart grid systems.

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